MetaMed: Enhancing Medical Education through Immersive Virtual Reality

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1. Introduction

Introduction

- A VR-based medical training platform.
- Develop VR medical simulations with haptic feedback, enabling realistic training for medical students.
- Reduce costs and ethical concerns associated with traditional surgical training methods.
- Unity 3D and Blender for environments.

2. Execution Plan

Execution Plan

FYP-1

- Environment setup in Unity3D and Blende
- Assets of App (Patient body, bed, tables and environment).
- Developed basic animations in 3D.

FYP-2

- Environment rendering to HDR.
- Rendering of Assets .
- Handling of Assets(Syringes, Pill Bottle, Sanitizers).
- Integration in oculus.
- Testing in oculus set.

3. UML Diagrams

Use case Diagram

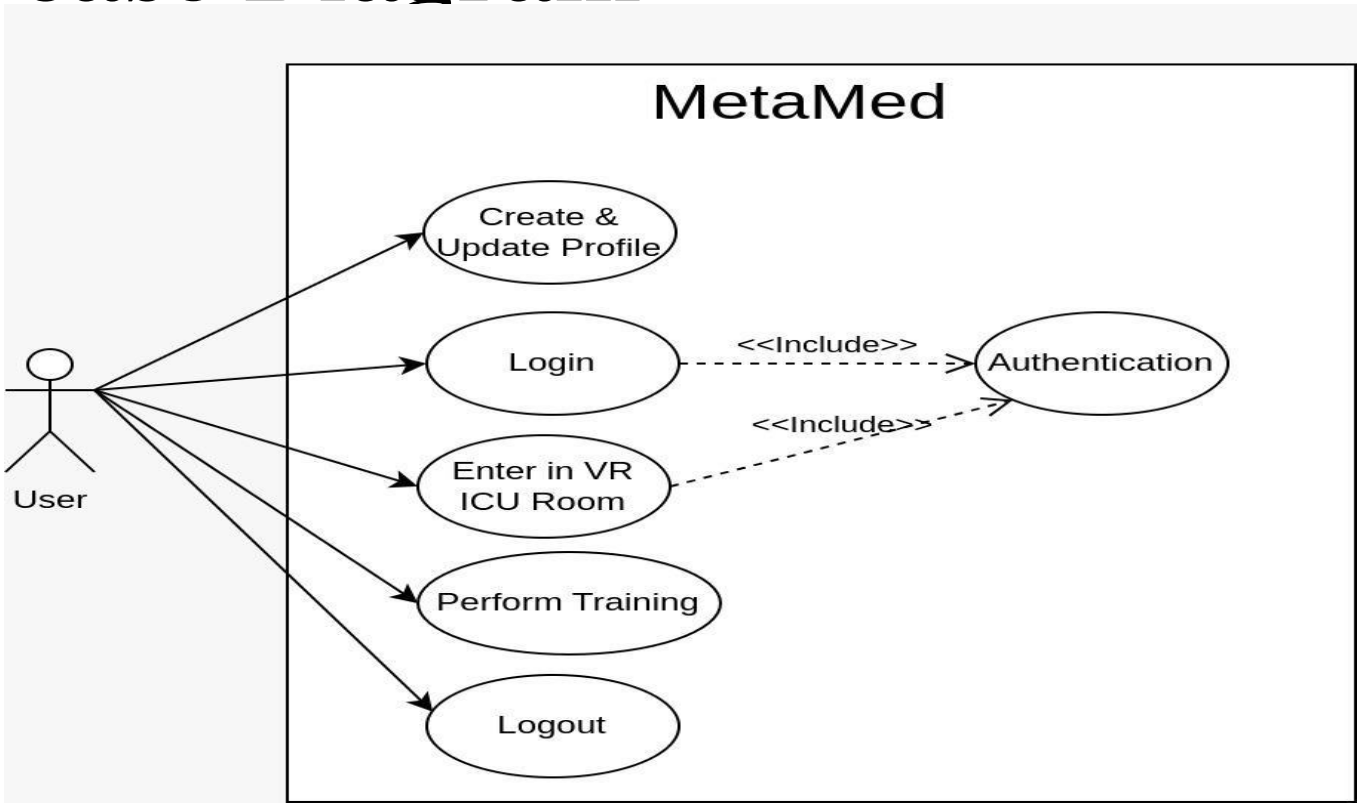

Fig.1.0. Use Case Diagram

System Diagram

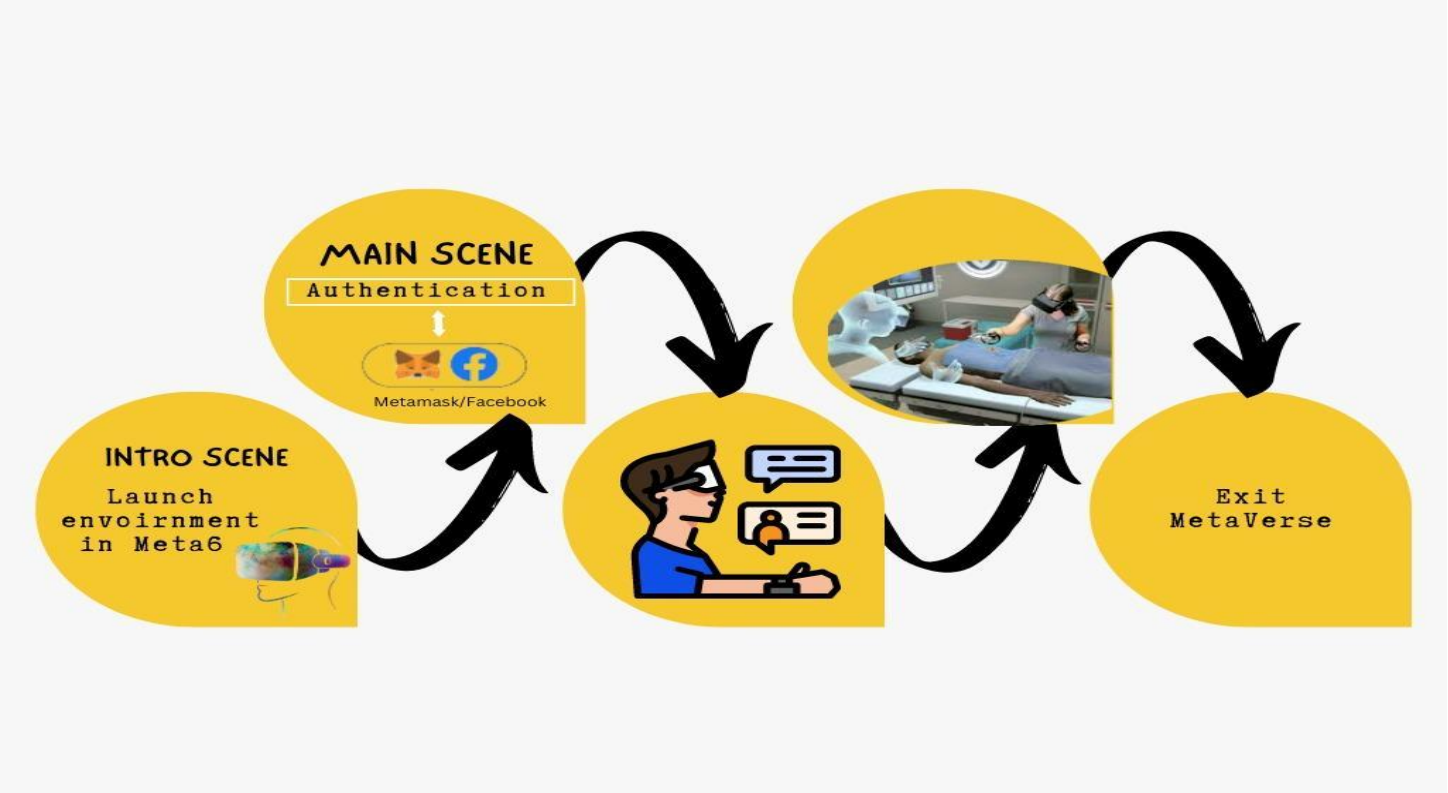

Fig.2.0. System Diagram

Activity Diagram

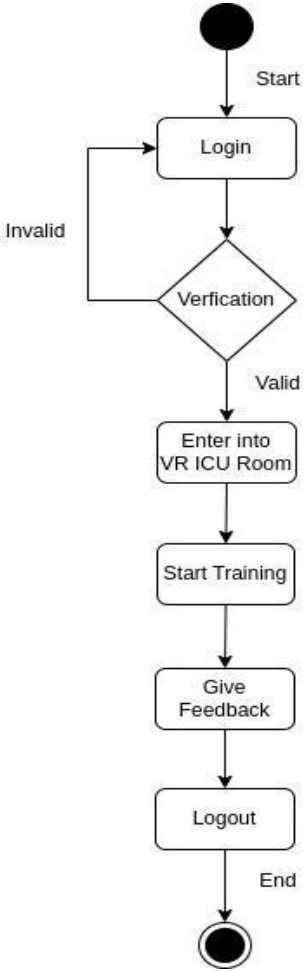

Fig.3.0. Activity Diagram

4. Tools Used

- Blender
- Unity 3D
- VR Headset
- Microsoft Visual Studio
- C#
- VRTK

5. App Highlights

Patient Room

Fig.4.0. Patient Room

Tools Area

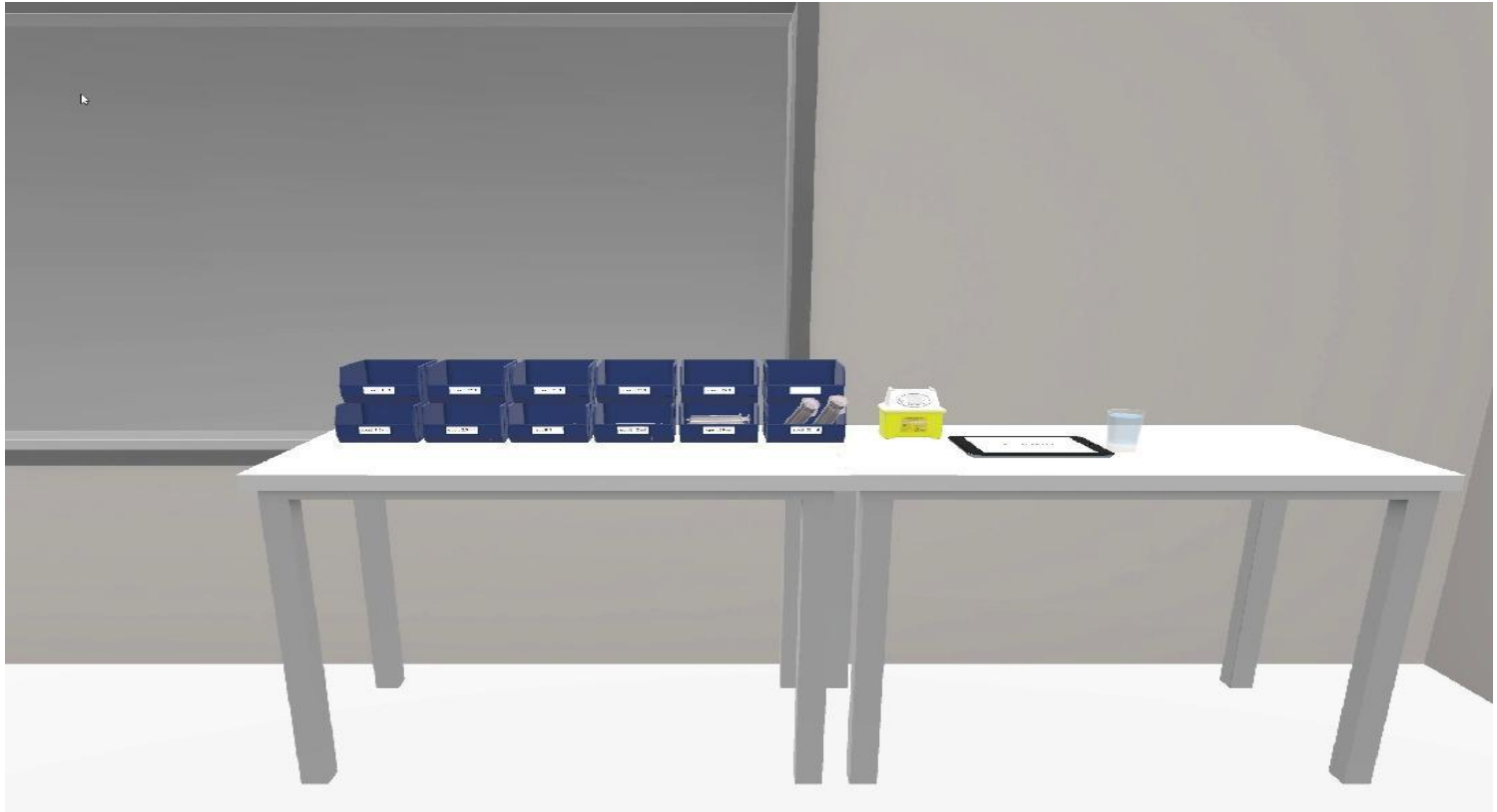

Fig.5.0. Tools Area

6. Teamwork

✓ Done
✓ Inprogress

FYP- 2	DAWOOD	WALEED	SHOAIB
REPORT		✓	✓
ASSETS DEVELOPMENT	✓		✓
SIMULATION DEVELOPMENT	✓	✓	
TESTING FUNCTIONALITIES	✓	✓	
INTEGERATION	✓		✓
SLIDES		✓	✓

Demo Video